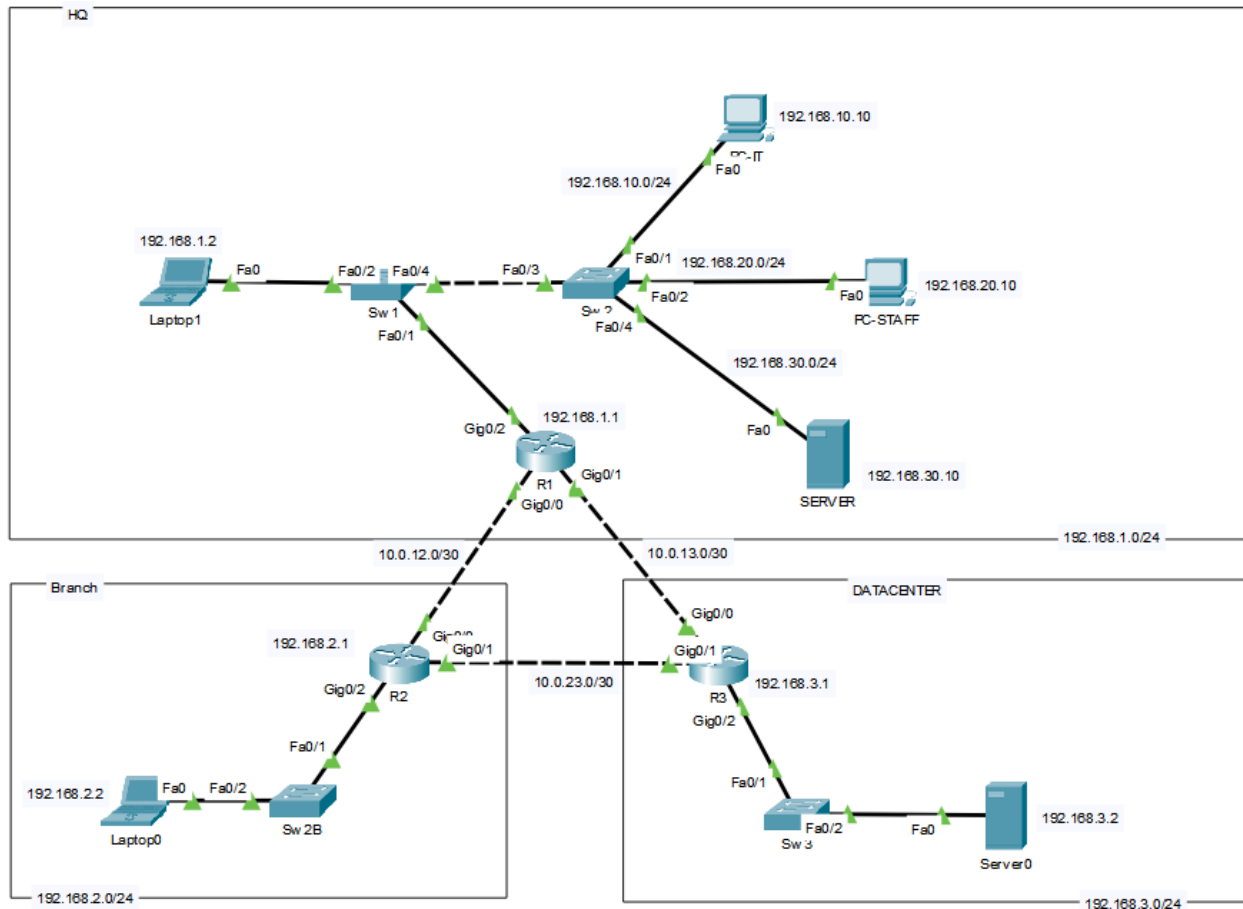




Cấu hình OSPF

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.2.1	1	FULL/BDR	00:00:39	10.0.12.2	GigabitEthernet0/0
192.168.3.1	1	FULL/BDR	00:00:30	10.0.13.2	GigabitEthernet0/1

```

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/0
L    10.0.12.1/32 is directly connected, GigabitEthernet0/0
C    10.0.13.0/30 is directly connected, GigabitEthernet0/1
L    10.0.13.1/32 is directly connected, GigabitEthernet0/1
O    10.0.23.0/30 [110/2] via 10.0.12.2, 00:00:01, GigabitEthernet0/0
      [110/2] via 10.0.13.2, 00:00:01, GigabitEthernet0/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet0/2
O    192.168.2.0/24 [110/2] via 10.0.12.2, 00:00:01, GigabitEthernet0/0
O    192.168.3.0/24 [110/2] via 10.0.13.2, 00:00:01, GigabitEthernet0/1

```

Kiểm tra ping LAN từ R1 qua R2 và R3

```
Router#ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

Kiểm tra đường đi thông qua tracerouter

R1 → R2 và R3

```
Tracing the route to 192.168.3.1

  1  10.0.13.2      0 msec    0 msec    0 msec
R1#
R1#traceroute 192.168.2.1
Type escape sequence to abort.
Tracing the route to 192.168.2.1

  1  10.0.12.2      0 msec    0 msec    0 msec
```

R2 → R3

```
R2#traceroute 192.168.3.1
Type escape sequence to abort.
Tracing the route to 192.168.3.1

  1  10.0.23.2      0 msec    0 msec    0 msec
```

Ping từ laptop Branch

Laptop0	
☒ DHCP	☐ Static
IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	8.8.8.8

```
Laptop0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=9ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

Ping Từ laptop HQ

Laptop1

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
IPv6 Configuration	

Laptop1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=9ms TTL=126
Reply from 192.168.2.2: bytes=32 time=5ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=4ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

Ping từ Server datacenter

Server0

Physical	Config	Services	Desktop	Programming	Attributes
IP Configuration					
IP Configuration					
☒ DHCP ☐ Static					
IPv4 Address		192.168.3.2			
Subnet Mask		255.255.255.0			
Default Gateway		192.168.3.1			
DNS Server		8.8.8.8			

```
Server0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Cấu hình acl chặn từ R2 ⇔ R3

R2: cấu hình acl chặn trực tiếp trên R2 chiều in từ g0/2

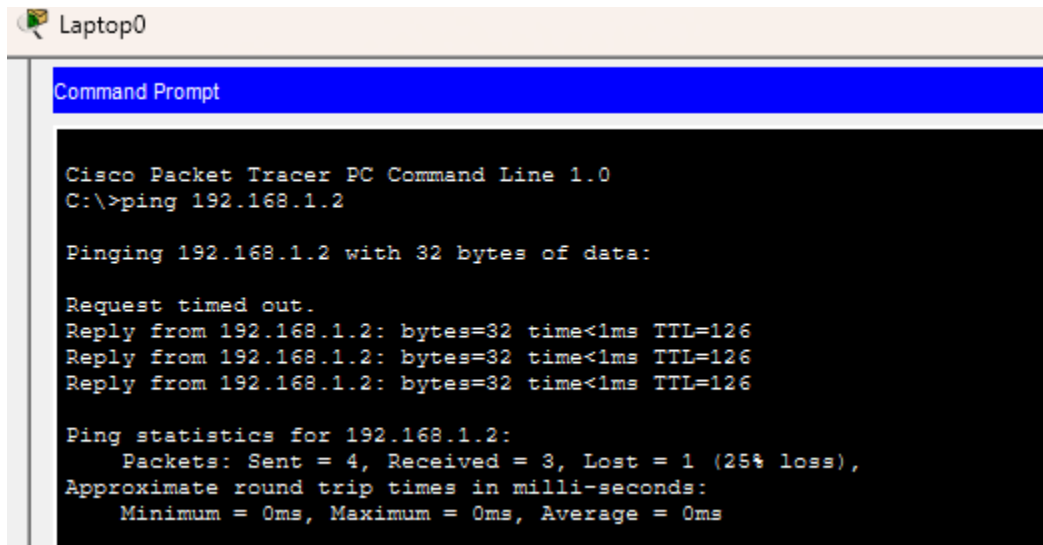
```
R2(config)#access-list 102 deny ip 192.168.2.0 0.0.0.255 192.168.3.0 0.0.0.255
R2(config)#access
R2(config)#access-list 102 perm
R2(config)#access-list 102 permit ip any any
R2(config)#inte
R2(config)#interface g
R2(config)#interface gigabitEthernet 0/2
R2(config-if)#ip acc
R2(config-if)#ip access-group 102 in
R2(config-if)#end
```

R3: cấu hình acl chặn trực tiếp trên R2 chiều in từ g0/2

```
R3(config)#access-list 103 deny ip 192.168.3.0 0.0.0.255 192.168.2.0 0.0.0.255
R3(config)#acc
R3(config)#access-list 103 per
R3(config)#access-list 103 permit ip any any
R3(config)#inter
R3(config)#interface g
R3(config)#interface gigabitEthernet 0/2
R3(config-if)#ip
R3(config-if)#ip acce
R3(config-if)#ip access-group 103 in
R3(config-if)#end
```

Test acl

Ping R2 qua R1



```
Command Prompt

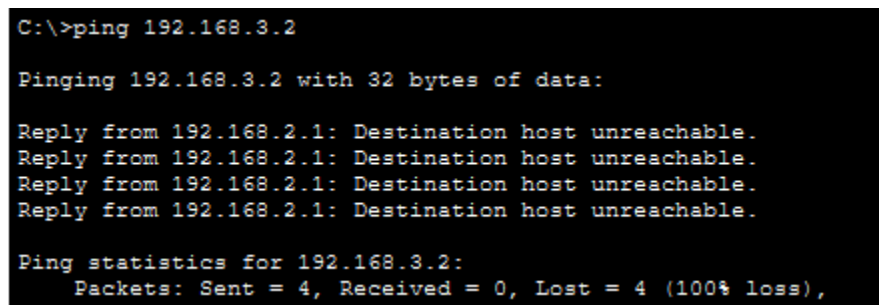
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping R2 qua R3



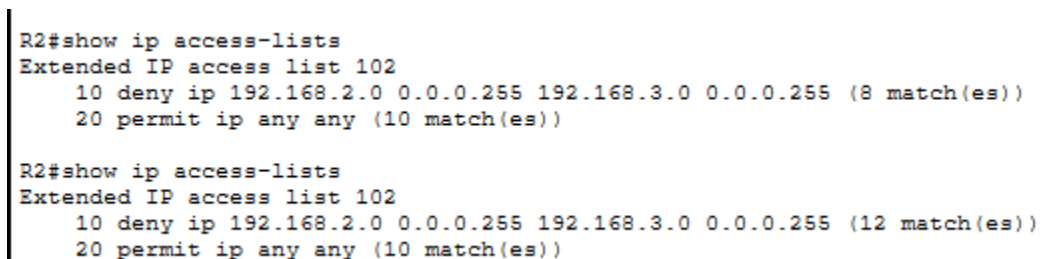
```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

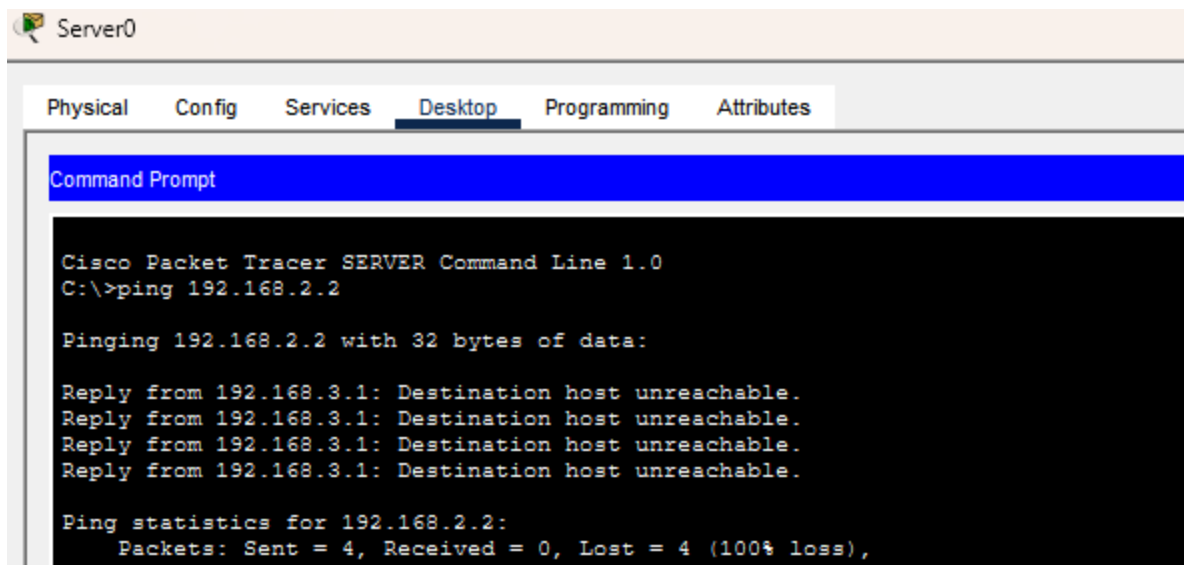
Xem log chặn trên R2



```
R2#show ip access-lists
Extended IP access list 102
 10 deny ip 192.168.2.0 0.0.0.255 192.168.3.0 0.0.0.255 (8 match(es))
 20 permit ip any any (10 match(es))

R2#show ip access-lists
Extended IP access list 102
 10 deny ip 192.168.2.0 0.0.0.255 192.168.3.0 0.0.0.255 (12 match(es))
 20 permit ip any any (10 match(es))
```

Ping test từ R3 => R2



R3 => R1

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Xem chặn trên R3

```
Extended IP access list 103
 10 deny ip 192.168.3.0 0.0.0.255 192.168.2.0 0.0.0.255 (4 match(es))
 20 permit ip any any (6 match(es))
```

Cấu hình VLAN & Trunk

SW1

SW1#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	IT	active	
20	STAFF	active	
30	SERVER	active	

SW1#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1
Fa0/4	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005
Fa0/4	1-1005

Port	Vlans allowed and active in management domain
Fa0/1	1,10,20,30
Fa0/4	1,10,20,30

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	1,10,20,30
Fa0/4	1,10,20,30

SW2

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	IT	active	Fa0/1
20	STAFF	active	Fa0/2
30	SERVER	active	Fa0/4

SW2#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/3	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/3	1-1005

Port	Vlans allowed and active in management domain
Fa0/3	1,10,20,30

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/3	1,10,20,30

Router-on-a-Stick

show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.12.1	YES	manual	up	up
GigabitEthernet0/1	10.0.13.1	YES	manual	up	up
GigabitEthernet0/2	192.168.1.1	YES	manual	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Ping test

PC IT 192.168.10.10

```

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=8ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

```

PC STAFF 192.168.20.10

```

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

SERVER 192.168.30.10

```

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=8ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

```

ACL chặn STAFF / SERVER

```

R1#show acce
R1#show access-lists
Extended IP access list 110
  10 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
  20 permit ip any any

```

Ping test STAFF tới SERVER

```

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

```

R1#show access-lists
Extended IP access list 110
  10 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
  20 permit ip any any (4 match(es))

```

VLAN IT vẫn truy cập server bình thường

```
PC-IT
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

! Nhưng cách này chặn luôn cả chiều vlan 30 => 20

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Cách tối ưu hơn cho cho phép chặn 1 chiều từ vlan 20 => 30

```
R1(config)#access-list 110 deny icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 echo
R1(config)#access-list 110 permit ip any any
R1(config)#interface gigabitEthernet 0/2.20
R1(config-subif)#ip access-group 110 in
R1(config-subif)#end
```

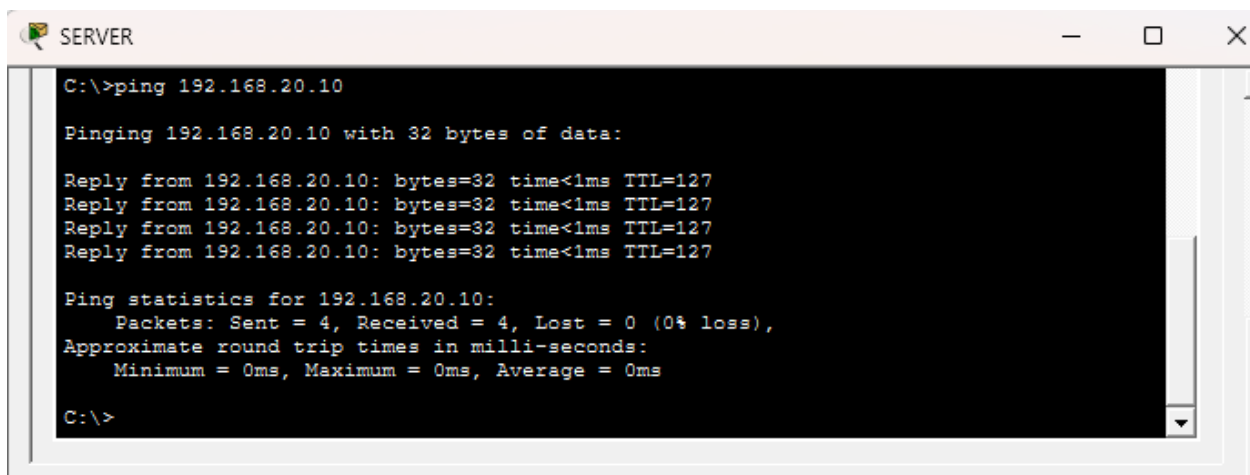
Test lại trên 2 vlan

```
PC-STAFF
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```



```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
show access-lists
Extended IP access list 110
 10 deny icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 echo (4 match(es))
 20 permit ip any any (4 match(es))

R1#
```
